

Project 2 - Group 4

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Abstract

This report will go over the development of a decision-making artificial intelligence model which is innately explainable. The model achieves this explainability by traversing a goal tree and making decisions based on the user's preference and predefined social norms. Finally, the model creates a natural language explanation for the user to understand its steps during the decision-making process. Ultimately, the natural language explanations were submitted for peer evaluation with a Likert scale form to assess the model performance. Based on the form's results the model's explanations performed decently.

Keywords

XAI, goal-trees, natural language explanation, explanations, Likert questionnaire

1 Introduction

As the popularity of Artificial Intelligence increases, more and more people have integrated it into their daily lives. However, most people place blind trust in Artificial Intelligence without understanding how it works or makes decisions. This paper aims to create a model that can go through goal trees, make decisions based on user preferences and social norms, and give a natural language explanation of the decision-making process to help the user understand. The initial step was to create an algorithm that enumerates execution traces from a given goal tree while annotating decisions based on the given social norms. Then, the decision-making is optimized by the user-specified preferences. Together, these algorithms ensure that our model considers both the societal norms and preferences of the user at the same time and integrates them into the decision process. The second step was to create natural language explanations for the decisions that the algorithm provided. This way, users can better understand how the algorithm makes its decision. These steps will be explained in greater detail in their corresponding sections. Finally, a peer evaluation was conducted on a Likert scale questionnaire with questions related to the quality of the generated explanations. Section 2 will explain the algorithm used to provide natural language explanations to the user. Then, Section 3 will describe the conducted experiments, the testing process, and its data collection and provide an analysis and discussion of these results. Lastly, Section 4 will give an overall conclusion.

2 Algorithm for Generating Natural Language Explanations

The algorithm uses an execution trace to generate the explanation of an action that the agent performed. It maps the information from the execution trace such as the agent's actions and the variables for said actions into human-understandable language. This execution trace contains every relevant information about the factors that affect the action to explain. The natural language algorithm loops over these factors and explains them in order. The way these factors effect the output explanation is shown in detail with the pseudo code Algorithm 1. These factors are:

- **Goal (D):** The goal of the agent.
- **Beliefs - Valid (C):** The condition under which the action is possible.
- **Beliefs - Failed (F):** The condition under which the action is not possible.
- **Preferences (V):** The agent's priorities for selecting actions.
- **Norms - Obligations (O):** The required actions that the agent must follow.
- **Norms - Prohibitions (P):** The prevented actions that the agent must avoid.

The algorithm initially describes the goal of the agent and the action taken to achieve this goal. This is achieved by using factors *D*, the agent goal, and *C*, the agent beliefs under which achieving the goal is possible, as shown in the Algorithm 1. Then, the algorithm explains if the pre-conditions for taking the action are satisfied by the agent's beliefs by utilizing the *C* factor and the *P* factor, prohibited actions that the agent must avoid. After this is done, the algorithm explains the preferences and norms that are provided to the agent and how they fit into the decision-making process. This is done via the use of factors *V* for preferences and *P* and *O* for the prohibited and required norms. Later, the actions that could not be taken are explained with their corresponding failed condition using the factor *F*, agent beliefs under which a specific action is impossible. Finally, the goal and the remaining actions necessary to achieve the goal are mentioned in the last sentence.

Overall, the algorithm considers a large variety of possible explanations and returns the most suitable narrative considering the conditions, beliefs, preferences, actions and norms of the user.

Algorithm 1 Generate Natural Language Explanation**Require:** *explanation_factors***Ensure:** *final_summary*

Initialize sentences as a dictionary with keys C, N, F, P, L, D, V, U and empty lists.

Initialize variables: *ordered_prefs*, *main_goal*, *chosen_option*, *precondition_in_C*, *final_summary*, *norm_violations*, *precondition_factors*, *norm_type*, *norm_raw*.

if *explanation_factors* is empty **then**

return action-prohibition clash explanation

else

for all *factor* $\in$ *explanation_factors* **do**

if *factor.type* = 'D' **then**

main_goal $\leftarrow$ 'get a coffee' **if** *factor.value* = 'getCoffee' **else** *chosen_option* $\leftarrow$ READABLEACTION(*factor*[1])

else if *factor.type* = 'C' **then**

chosen_option, *precondition_in_C* $\leftarrow$ READABLECONDITION(*factor*[1], *factor*[2])

else if *typ* = 'U' **then**

ordered_prefs $\leftarrow$ [*pref_names*[*i*] for *i* in *pref_order*]

else if *factor.type* = 'V' **then**

 Append READABLEACTION(*factor*[1]) + READABLEACTION(*factor*[4]) to *sentences*[V]

else if *factor.type* = 'N' **then**

 Append READABLEACTION(*factor*[1]) to *norm_violations*

else if *typ* = 'F' **then**

 Append READABLEACTION(*factor*[1]) + READABLECONDITION(*factor*[2]) to *sentences*[F]

else if *factor.type* = 'P' **then**

 READABLEACTION(*factor*[1]) + READABLEACTION(*factor*[2]) to *precondition_factors*

else if *typ* = 'L' **then**

 Append READABLEACTION(*factor*[1]) + READABLEACTION(*factor*[3]) to *sentences*[L]

end if

end for

if *main_goal* + *chosen_option* **then**

 Append *main_goal* + *chosen_option* to *final_summary*

end if

if *precondition_in_C* **then**

 Append *precondition_in_C* to *final_summary*

end if

 Append *precondition_factors* to *final_summary*

if *ordered_prefs* **then**

 Append preference summary to *final_summary*

end if

if *norm_violations* is not empty **then**

if *norm_type* = "P" **then**

 Explain the violation of prohibition

if *norm_type* = "O" **then**

 Explain obligation violation

end if

end if

 Append all V, F, and L sentences to *final_summary*

return *final_summary*

3 Experiments

This section will give a detailed explanation of the experiments conducted to evaluate the algorithm outlined in Section 2 in Section 3.1, then present the corresponding results in Section 3.2 and finally discuss the presented results in Section 3.3.

3.1 Experimentation Setup

To evaluate the algorithm presented in Section 2, a user survey was conducted using FeedbackFruits¹. The survey was generated by the professor and the participants were randomly allocated students from the Explainable AI course at Utrecht University.

Participants were asked to evaluate 5 different natural language explanations that were generated using the previously described algorithm and a predefined input. For each explanation, participants were asked to evaluate 6 questions (seen in Table 1) on a 5-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree). The users were provided both a description of a scenario (following a pre-defined template) and the corresponding explanations.

Each of the questions provided is used to evaluate different criteria for human explanations, which can be seen in Table 1.

Question Number	User question	Explanation Criteria
Q1	The explanation clearly shows how the system arrived at its decision.	Transparency
Q2	The explanation covers all relevant information correctly.	Completeness, Correctness
Q3	The level of detail in the explanation is sufficient for me to understand the decision.	Sufficiency, Informativeness
Q4	The explanation appropriately addresses the question.	Relevance
Q5	I believe the system is making decisions in my best interest considering my preferences.	Alignment with User Preferences
Q6	I am satisfied with the explanation provided.	Satisfaction

Table 1: User questions and numbering

3.2 Results

A total of 12 users provided their responses to the survey, of which 8 participants provided full ratings for all input files. The total number of comments left for the data analysis was 64; however, not all could be connected to a certain input.

A Likert plot for the overall rating for each question can be seen in Table 1 while a summary of the mean, mode, and median values of the evaluations for each question is shown in Table 2.

Please note that in the questionnaire a Likert scale evaluation was administered and completed by 10 participants. However, due to the structure of the questionnaire, it is unclear what documents this

¹<https://feedbackfruits.com>

	Q1	Q2	Q3	Q4	Q5	Q6
Median	4	4	4	5	5	4
Mean	3.72	4.02	3.87	4.35	4.2	3.76
Mode	4	4	4	5	5	3

Table 2: Summary of Overall Ratings

rating corresponds to, and thus will be omitted from the analysis. For completeness sake, these results are still provided in Appendix A.

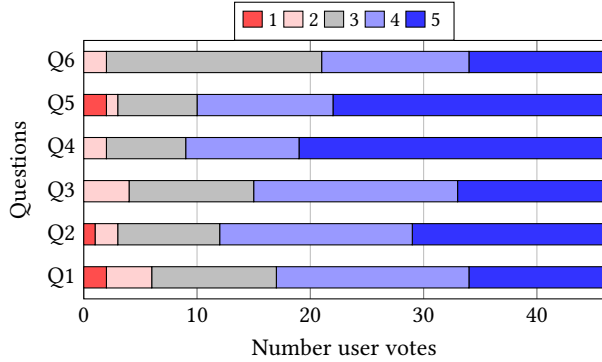


Figure 1: Stacked Likert Ratings for All Inputs

It can be seen that ratings of 1 or 2 are the least common answers, the most common being 4 and 5. This is also shown in the mean and mode values. Overall, this shows that the provided natural language explanations were satisfactory to the users, but there are some areas in which they can be improved.

This is further supported by the comments provided by the users, where many users mentioned the need for grammar improvement and easier flow of the natural language, and focusing more on only important information to the user in the given explanation. Positively users remarked that the explanations were easy to follow, clear, and concise.

Through the mean, mode, and median values one can see that the areas with the most need for improvement are those tested through Q1, Q6, and Q3, more specifically the following criteria: Transparency, Satisfaction, and Sufficiency.

3.3 Discussion

As mentioned in previous sections, the decision-making model uses natural language to generate explanations for its decisions. These explanations were evaluated by peer feedback in a Likert scale form. Firstly, one of the key observations from the user feedback is that the generated natural language explanation could be improved in terms of fluency and narrative coherence. The feedback shows that, while the model was informative, the information was fragmented too much, which affected the flow and clarity of the explanation, making it hard for users to understand and piece information together. The feedback also pointed out that the structure of the given explanation and its reasoning made it hard to comprehend how the model came up with its decisions.

Another common problem was that the feedback suggests that repeating norms and preferences is redundant. Finally, some of the criticism proposed that the tone of the explanation was too technical, which made it hard to follow. The suggestion was to implement a more realistic and relatable narrative. However, a concern is that this might induce unnecessary text and overcrowd or complicate the explanations even more. Overall, the explanations were found satisfactory by the users and helped their understanding of the process, as seen in both the commentary and Likert scale data. As seen by these remarks and comments, the natural explanations could be improved to achieve a more fluent, consistent, and human-like explanation.

In the future, the mentioned problems could be improved on to achieve a better structured and more natural-sounding explanation. The priority improvement for better user understanding of the agent's actions would be restructuring the order in which the actions and the reasons for those actions are presented, ensuring they follow a coherent and logical sequence. It is also logical to remove the redundant information about the preferences and the norms that the agent considers, as this disrupts the flow of the explanations and adds unnecessary complexity. This is undesirable, as users mostly prefer explanations that are brief and simple. Furthermore, the use of simpler language with less technical terms would help make the explanations accessible to a broader range of users. Finally, to improve upon the experimental setup, feedback could be received from a more diverse crowd to see how the explanation appeals to different groups of people.

4 Conclusion

To conclude, this study focuses on a decision-making model and its explainability using natural language explanations. This is an essential method for achieving more transparent and understandable AI models. Although the presented model can traverse the goal tree and come up with natural language explanations and was found to be satisfactory, the peer feedback conveyed some areas where the model could be improved such as clarity and structure of explanations. This suggests that the explanations should be given in a clearer, more natural and structured manner to help the user understand and provide them with a better user experience. Future work should focus on enhancing these kinds of problems and implementing these explanations to not only decision-specific models but a variety of AI models. Additionally, gathering more feedback from a more diverse group could reveal more about what to improve in the explanations to make them more accessible and informative. With these kinds of improvements, the model should be able to connect complex algorithmic decision-making and user-friendly, transparent communication.

A Likert result from questionnaire

Table 3 shows the results of the user evaluations collected in the questionnaire that, due to the configuration of the questionnaire, cannot be connected to a specific input and are therefore excluded from the data analysis. These results are recordings from 10 different users.

Likert Rating	Q1	Q2	Q3	Q4	Q5
5	2	1	1	4	6
4	5	6	4	5	1
3	2	3	5	1	2
2	0	0	0	0	0
1	0	0	0	0	1

Table 3: Likert Ratings from Questionnaire that do not correspond to any specific input

B User Comments

Table 4 shows a subset of the received comments for better oversight only the comments from users that provided a general comment for each of the 5 tested inputs are provided.

Table 4: General comments left by users for a specific input

User	Input	Comment
User 1	1	"The phrasing in the explanation could be improved to say something like ""The action was not performed..." instead of "" The action cannot be explained..."". Because with later there is an implication that the agent is not capable of explaining itself."
User 2	1	The explanation is correct, but it does not include the logic of the chosen trace.
User 3	1	"the explanation correctly says that the action couldn't be executed but it should analyze how the agent achieved the goal of having a coffee as it is mentioned in the steps ""following actions"" and its the main goal"
User 4	1	"The explanation is contradictory—if paying at the shop is prohibited, it should clarify why the agent still chose it"
User 5	1	"The explanation is clear, correct and concise. That the action of paying at the shop is prohibited is important relevant information and is mentioned appropriately."
User 1	2	"The explanation's structure makes it difficult to follow the flow of thought. In some places, it seems very unnatural and robotic. The reason why the other two places were discarded is only given at the end of the explanation, which is not ideal."
User 2	2	"The explanation lacks a natural flow, jumping between statements without clear connectivity. Additionally, the reference to ""the agent"" could have been avoided by writing in the first-person perspective, which would have made the explanation more cohesive."
User 3	2	"the explanation is a bit complicated, it mistakenly says about making coffee in the kitchen which is not mentioned. i would expect a better flow and not answer every option if it satisfied or not as they are questions and it doesnt have to do with the explanation but i think that the actions should be more and not just going to kitchen and getting coffee, it should have about the card"
User 4	2	"While it describes why the kitchen was chosen, there's confusion around preconditions for using one's own card and the incorrect statements about "colleague is available" or "Ann is in the office" not being satisfied. It references user preferences but doesn't show actual cost comparisons."
User 5	2	"The explanation is correct, but the action node names are not expressed correctly (in a grammatical/syntactic way) which makes reading this explanation a bit harder compared to if it was written as normal phrases (e.g. because it didn't have money instead of ""because it has money was not satisfied""). The phrasing of the preferences could also imply that price and time are on equal grounding when it's actually price > time, so this could give the use the impression that the agent does not have their best interests in mind. Additionally, the order of the different explanatory components is quite confusing and makes the explanation difficult to understand. It has to check that it has its own card, but then the explanation suddenly turns to the preferences of the user, which had nothing to do with the sentences before and after this phrase about preferences. Lastly, the last two phrases pertaining to why the agent didn't get coffee from the shop or from Ann's office also seem misplaced, because this was a decision that was made at the beginning (choosing get getting kitchen coffee over getting coffee from Ann's office or the shop)."
User 1	3	"The explanation is hard to follow. It doesn't follow a step-by-step explanation, in some places it describes actions from the result to the initial state, and in others - follows a more top-down approach starting from the initial state. Focusing on one of the approaches can greatly improve the feeling of the explanation."
User 2	3	"The explanation is generally fine but lacks smooth flow. Additionally, the statement about cost could be phrased more clearly for an end user: ""The agent preferred to pick up the coffee from the shop over getting coffee from Ann's office because it had a lower overall cost, especially with quality as the top priority."" The logic behind ""lower cost is better"" should be better explained for an end user who may not fully understand this reasoning."
User 3	3	"The explanation is good, but some comments that i think it should have to be more on target and follows all requirements are: -it is not mentioned that should check that is at the shop. -quality was a priority as you mentioned but shop is more expensive than ann's office coffee. so saying that has lower overall cost is wrong. -it needs a better overall flow to looks natural and not like answering if questions are satisfied"
User 3 (2)	3	"the explanation is a bit complicated, it says that prioritize quality but ann's Office coffee is not more expensive (as it says that the shop has a lower cost). instead of answering as ""The agent"" it should say ""i"" as the agent is speaking. the flow needs improvement and not answering if it is satisfied or not. Also here the followed actions should be more"
User 4	3	"It does a decent job explaining how the shop option was chosen and mentions quality as the top priority. However, it's slightly unclear why the agent says "staff card is available was not satisfied" for the kitchen route, which is unnatural. Overall, it's coherent."

User	Input	Comment
User 5	3	"The explanation is correct and compared to input 2, most action node names are integrating into the text in a natural way. However, the ordering/phrasing of the different components is once again not the most straightforward which makes reading the explanation confusing. ""Before being able to pay at the shop, it was necessary to check that it has money"" - while this action makes sense, it was mentioned before that a prior decision was made because it had money. But rather, this decision was possible because it had money, and after checking whether it has money, there is not confirmation from the agent that it has: the explanation proceeds to discuss the preferences. It would have been clearer if the explanation also included this confirmation of the agent having money, or not mentioning this phrase about money again, since it was already mentioned it had money in the second sentence. One of the reasons why the agent chose shop coffee, which is because coffee from the kitchen wasn't possible, is mentioned last again, which also makes the explanation less clear."
User 6	3	Seems like a good explanation! Some room for grammar improvements. And i would not talk about costs if it not refers to money.
User 1	4	"- The explanation is hard to follow. - Your agent says ""This decision was made because it has money"", but in the very next sentence it says ""it was necessary to check that it has money"". This line of thought is unclear and the order might confuse people. - ""...it had a lower overall cost, especially with quality..." - how can you explain what the cost of quality is?"
User 2	4	"The explanation is generally fine but lacks smooth flow. Additionally, the statement about cost could be phrased more clearly for an end user: ""The agent preferred to pick up the coffee from the shop over getting coffee from the kitchen because it had a lower overall cost, especially with quality as the top priority."" The logic behind ""lower cost is better"" should be explained more clearly for an end user who may not fully understand this reasoning. In natural terms, the shop has a higher monetary cost but offers higher quality coffee, which the user prioritized. This could have been better clarified in the explanation."
User 3	4	"the explanation is good, has good reasoning, only comments that shop has higher cost and not lower and that it should include all actions (GetCoffee)"
User 4	4	It's decent overall
User 5	4	"The explanation is quite clear and the structure also makes sense. The information is correct and relevant, and overall the explanation is phrased in an easy to follow manner. It includes the preferences correctly, giving the user the impression that the agent has their best interests in mind. The only component that is missing is why it didn't choose to get kitchen coffee: all the preconditions for that trace were present. This could have made the explanation complete."
User 6	4	"Seems not that intuitive to an end user as cost not implies money."
User 1	5	"The explanation is a bit long and it is hard to follow the line of thought. It is explained why we choose the kitchen coffee over the shop coffee, however reasoning about Ann's office is given only after explaining that we are getting our own card. This order makes it hard to follow."
User 2	5	"The explanation is generally correct but lacks a natural, human tone. For example, the phrase ""Ann's office because Ann is in the office was not satisfied"" feels awkward and could be rephrased for clarity and flow"
User 3	5	"the explanation is efficient in terms of reasoning but it need some refinement in: -there is no prohibition about going to shop but paying in shop. - borrowing card from another colleague doesn't affect price but time - i would like a more natural language explanation, agent should talk as ""i"" and not the agent and also to have a flow and not answer if everything is satisfied or not"
User 4	5	"The explanation covers why the shop path is off-limits and why the kitchen route was chosen, but it's confusing about ""Ann is in the office was not satisfied." That sounds contradictory. Also, it doesn't clearly show how price is measured or how going to the kitchen really won out on cost."
User 5	5	"The explanation is correct and covers the relevant information. The structure is still not entirely clear, with mentioning the path to Ann's office not being possible almost at the end of the explanation. Additionally, the sentence ""Before being able to go get their own card, it was necessary to check that it has their own card"" is confusing because it seems to imply the same thing twice, because by trying to get your own card, you will find out if you have it."